BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

LESSON 077 · Lesson 077 · Linear Search: Code and Trace

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Can linear search work with unsorted input?

Q2. What is not-found sentinel?

Q3. What does O(n) time mean?

Q4. Array a=[6,2,9,4]: set a[2]=a[0]+a[1]. Give updated array and total.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

linear search — Sequentially compare elements.

sentinel — Special result indicating a condition.

index — Zero-based position.

New learning

Understand — Visit each array value in order; linear search does not require sorted data.

Trace — Return the first matching index or a distinct not-found value such as −1.

Explain — Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Worked example

Find 6 in unsorted [1,5,6,3].

1. Check the working — Check indexes0,1,2, return2; stop before3.

We Do · Practise with guidance

Find 8 in [3,7,2,9].

```
No equal item, so return−1 after four checks.
```

Check the working

No equal item, so return−1 after four checks.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace 2 in [4,9,2,7], give checks and return. [6]

Question 2

Write C# linear search returning index/−1. [6]

Question 3

State linear search best/worst time, space and one edge test. [6]

Exit Ticket · Answer before checking

Linear search requires a pre-sorted array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Yes.

Q2. Distinct return such as −1.

Q3. Work grows linearly with n.

Q4. [6,2,8,4], sum20.

You Do pages · Independent questions

Q1. index0=4 [1], index1=9 [1], index2=2 [2], return2 [2].

Q2. Loop 0..Length−1 [2], check equality [2], return i [1], return−1 after loop [1].

Q3. Best first item O(1) [1], worst absent O(n) [2], auxiliary O(1) [1], empty array returns−1 [2].

Exit Ticket · Model answer

False. No sorting is needed.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: